# XAI In Healthcare With Deep Learning For Medical Image Diagnosis

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***Abstract*:AI and DL are revolutionizing healthcare as they have led to the automated analysis of medical images. The use of deep learning algorithms has resulted in monumental success in detecting diseases like Pneumonia, Tuberculosis, and Skin Cancer in medical images such as X-rays. Although they have proved to be accurate diagnosis tools in healthcare, all of them can be stated to be black boxes in reality as it has become rather challenging to reference a particular prediction. Solutions like Grad-CAM of Explainable Artificial Intelligence have made it possible to have a clear view of the regions of interest being identified in references to model predictions.**

***Keywords: Artificial Intelligence, Deep Learning, Convolutional Neural Networks, Explainable Artificial Intelligence, Grad-CAM, Medical Image Analysis, Disease Detection, Pneumonia, Tuberculosis, Skin Cancer, Interpretability, Healthcare***

## I.INTRODUCTION

Artificial Intelligence (AI) has emerged as a revolutionary paradigm within the field of the medical sector over the past few years, especially due to its implementation in the medical image analysis field. Nevertheless, within the context of the various AI paradigms, Deep Learning algorithms have enabled the computer to automatically extract key features from the medical images, thus enhancing the efficiency and correctness in the early diagnosis and determination of the disease in the individual. Furthermore, within the context of the various Deep Learning algorithms, the Convolutional Neural Network algorithms have demonstrated unparalleled efficiency in the interpretation of Chest X-ray images, Computed Tomography images, and dermoscopy images in the determination of patients suffering from Pneumonia, Tuberculosis, and Skin Cancer respectively.

Moreover, in addition to the excellent efficiency within the medical image analysis technique and correct determination in the medical diagnostic analysis, the majority of these Deep Learning algorithms operate within the black box setup, wherein there is limited information provided on the efficiency and determination within the box setup. Therefore, in order to enable the elimination of aforementioned drawbacks, the new AI paradigm, referred to as the 'Explainable Artificial Intelligence' (XAI) has emerged in the past few years, in an aim to enable the increased transparency and interpretability within the AI algorithms.

Furthermore, within the context of the aforementioned algorithms of the 'Explainable Artificial Intelligence' algorithms, the 'Grad-CAM' technique, referred to as the 'Gradient-weighted Class Activation Map' technique, directly offers information within the context of the 'medical images' about the key 'Regions of Interest', which affect the AI algorithms in the determination in the 'medical image analysis' assessment. Thus, it has facilitated the increased acceptance in the AI algorithms within the 'medical diagnostic' process and analysis.

Thus, through the development and implementation of the 'Diagnostic Assistance Systems' within the 'Deep Learning' and 'Explainable Artificial Intelligence' paradigms, the medical research scholars aim to facilitate the development of accurate and 'interpretable' systems in the 'medical diagnostic' analysis, not only reliable but more importantly, within the 'sound ethical' principles in the 'medical diagnostic analysis' technique.

## II.LITERATURE SURVEY

deeplearning is presently a foundational technology in Computer Aided Diagnosis (CAD) systems for medical images, with ResNet, DenseNet, VGG, EfficientNet, and MobileNet being deep learning architectures used for state-of-the-art performance in X-ray, CT, MRI, and dermoscopy image analysis. Although these architectures have high accuracy in performance, they are black boxes, which restrict their usage in medical institutes. Present literature focuses on Explainable AI, which analyzes AI solutions to understand rationale.

Scope and Positioning. A type of survey literature charts the XAI literature in medical imaging, with methods being grouped based on application in tasks such as classification, detection, and segmentation. A primary focus is placed on ensuring a model's explanation is model-internally accurate and informative to medical end-users. Common architectures combine transfer learning using CNNs with an XAI explanation technique.

XAI Techniques:The most common XAI methods include saliency/attribution techniques such

•Grad-CAM/Grad-CAM++: class-specific heatmaps calculated using gradients leading to the last conv block, which have been largely used for lesion/region-ofinterest identification

.• LRP and Integrated Gradients: redistributes scores of a prediction into pixels/feature maps to assess contribution

.• LIME/SHAP: local model explanations based on a local perturbation or Shapley explanation, which can be used with a wide variety of models.Consensus in existing literature is: Grad-CAM gives interpretable spatial information, and LIME/SHAP can be used for instance-level model-agnostic explanations.

Datasets and Modalities,Dataset examples: Chest X-ray datasets for pneumonia, COVID-19, and tuberculosis, ISIC/HAM10000 for skin lesions, and datasets for brain MRI images Important considerations: appropriately curate datasets, balance on both classes, and preprocess all datasets in a standard manner in terms of resizing, normalizing, and augmentingCross-validation on different datasets is not common Evaluation MetricsEstimation of model performance is usually measured with Accuracy, Precision, Sensitivity/Recall, Specificity, F1-score, and AUC. Qualitative assessment is achieved through alignment of attention maps with regions indicated by a radiologist (overlap/IOU or rating).

Literature shows that incorporating XAI does not affect model accuracy negatively but can increase team accuracy with human and AI combination.

.Representative Findings

.CNN Baselines with ResNet/DenseNet/VGG architectures and Grad-CAM achieve a fixed accuracy of ~90-97% in specific tasks such as COVID-Xray, Pneumonia, and Brain Tumor MRI with meaningful Heatmaps LIME/SHAP increase local, instance-level interpretability with capabilities to interpret both correct and incorrect predictions.

Clinical studies and case analysis conclude that saliency maps focus attention better if heatmaps contain regions of interest and exclude random information.Limitations and Challenges;

Faithfulness vs. plausibility: "Visually pleasing maps may not always reflect causal model behavior."

Computation complexity: Perturbation methods, such as LIME/SHAP, introduce latency, which can be alleviated by

Public, normalized, numeric XAI evaluation metrics (such as those attained through annotation evaluation and counterfactuals).

Multimodal Explainability integrating imaging with EHR and labs to support decisions.

Prospective clinical studies assessing the impact of explanation on diagnostic accuracy, confidence, and time to decision.

Robustness & fairness audits where XAI assists in identifying and remediating biases over demographics & devices

Real-time, edge-computable XAI.

Synthesis. Looking across modalities and tasks, a recipe emerges in the literature: accuracy with fine-tuned CNNs, explanation with Grad-CAM, and instance-level explanation with LIME/SHAP—with evaluation using both discriminative metrics such as AUC and F1, and explanation-oriented evaluation. Such a combination will prove most effective in achieving a balance among accuracy, explanation, and usability, thus highlighting XAI as a pre-condition for sound, trustworthy, and accountable AI in healthcare.

## III.MATERIALS AND METHODOLOGY

### 3.1 Architecture

The system will integrate the DL and XAI techniques to automatically classify the disease through the medical images while ensuring that the process is transparent. The main process in the system will entail five steps: Data Input, Image Preprocessing, CNN Feature Extraction, Classification Layer, and Generate Explainable Output

It is a deep learning model that uses a framework incorporating Explainable Artificial Intelligence (XAI) for interpretable medical image diagnosis. Medical image datasets such as X-ray and dermoscopic images are obtained and preprocessed by resizing and normalizing the images and performing data augmentation for uniformity and consistency. The preprocessed images are then passed on to the Convolutional Neural Network (CNN) model following the CNN architecture design that involves layers of convolution, activation functions, and pooling for automatically extracting hierarchical image features related to the disease patterns. A classification output is obtained from the global average pooling layer and fully connected layer using the Softmax activation function for the predicted binary classifications and probabilities of disease outcomes. To increase model interpretability and explainability using the Grad-CAM model for medical image analysis, the model concentrates on the last convolutional layer by analyzing the output and providing image region maps related to the predictive model output and model decisions using the heatmaps for image region activation that contribute towards the predictive medical

image diagnosis model output and decisions. The training model is implemented using the cross-entropy loss function that is optimized using the Adam optimizer with model validation and evaluation using accuracy score, precision score, recall score, F1 score, and ROC-AUC score for assessing model efficiency and validation.

Data Input: Chest X-rays, CT scans, or dermoscopy images. This will be carried out using datasets other than the Chest X-ray dataset for Pneumonia, namely the Tuberculosis Dataset and Skin Cancer Dataset. Both datasets have images with both normal and abnormal instances for the same modality

2. Image Preprocessing

Image inputs are preprocessed for consistency and ensuring the image quality is maintained. This includes procedures such as the following:

• Fixed dimension resizing
• intensity values of pixels

DATA AUGMENTATION – Rotation, flip, and brightness changes.

These steps are very helpful for noise reduction and add a good degree of robustness to theCNN approach.

It comprises of automated extraction of deep hierarchical features in medical images using a CNN, such as ResNet18, MobileNetV2, or EfficientNetB0.

• The convolution layers identify spatial patterns like lesions and nodules.
• The pooling layers decrease the spatial dimensions and computation expense.
•Activation Functions: Use the ReLU function to provide non-linear capabilities for learning complex mappings.

This feature extractor will be done by a CNN trained on discriminative features between healthy and sick states

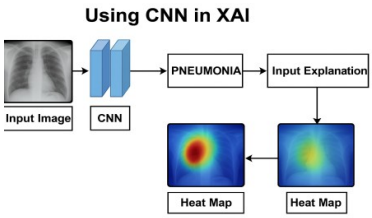


Fig 1: Architecture Diagram

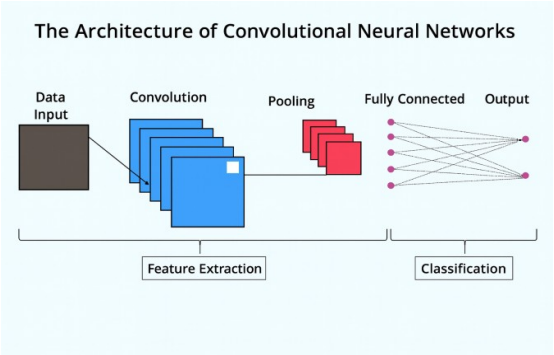


Fig 2:CNN Architecture Diagram

4. Classification Layer: The resultant features are then passed to the fully connected layers, which classify the images into predefined classes, including Normal, Pneumonia, Tuberculosis, and Skin Cancer, respectively.

• The output from the Softmax function is a probability score for every class.

The ultimate prediction is obtained by identifying the class with high probability.

Cross-Entropy Loss trains the network. The network is validated on the unseen data. Other metrics that can be used to evaluate the performance of the network include Accuracy, Precision, Recall, F1-Score, and AUC.

5. Interpretability Using GradCAM: GradCAM has been used to overcome the “black box” problem. Grad-CAM uses the gradients that the last convolutional layer receives in order to focus on the appropriate areas within the input image that affect the prediction.

• It creates heat maps of an image displaying attention areas.
• This would also help clinicians check that this model concentrated on biomedical areas of interest.

Therefore, not only will the system predict the disease but will also explain its choice visually.

6. Deployment and PDF Reporting on the Best-Trained Models: Integration of the best-trained models into a Flask web application. The user will be able to upload his or her X-ray images and choose the disease and the model he or she wants to use, and the result will show the prediction in real time. The system will create a PDF medical report

• Confidence score and Class by prediction

### 3.2 Algorithm Used In This Technology

#### 3.2.1.Convolutional Neural Network (CNN) — Classification

Learn visual features from medical images and output class probabilities (e.g., *Normal*, *Pneumonia*, *TB*, *Skin Cancer*.

**Forward pass per layer:**

**1.Convolution**

$$Y_c = \sum_{i=1}^{m}{}^{c} m_{C_w}(|u,v),\ W_{c,m}(u+j+v)\ b_c$$

where X is the input feature map, W is the kernel, bc is the bias, kernel size, c output channel.

**2.Activation (ReLU)**

$$ReLU(x) = \max(0, x)$$

**3.Batch Normalization**

$$x_m^x = = \frac{\mu}{\sigma^2}\ \sigma_i^2,\ \ y = s_c^{\varepsilon x} + \beta\varepsilon$$

**3.Pooling**

$$Y(i,j) = \frac{1}{W} = \frac{1}{1-\varepsilon}\ \hat{A}.$$

**4.Global Average Pooling (GAP)**

$$\alpha_{r\ c}^{C} = \frac{1}{HW}\ \xi_s\ \vee\ A_{ij}$$

**5.Linear + SoftmaxX**

$$\underline{s}_c = v_c\ \sqrt{\sum_{i=1}^{e} \frac{s\ c_{n,i}}{A_{ij}}}$$

$$p_c = \frac{e_c}{\sum e^{i\ of}}\ A_k(i,j)$$

**3.2.2. Training Objective & Optimizer**

Loss (Cross-Entropy for multi-class)

$$CL(\theta) = -\ \sum_{yc} y\ og p_c$$

with one-hot target y and predicted p.

**3.2.3.Grad-CAM — Explainability Algorithm**

Weights for each channel k in the last conv feature map A^k

$$\mathcal{L}_{\text{rad-CAM}} = \quad g_t = \frac{\partial A_x}{\partial_t + b_t},\ \ p_c\ \frac{\partial s_c\ A_{ik}}{\sum e_k e^{t'}}$$

**3.2.4.Evaluation Metrics**

To evaluate the effectiveness of the proposed Explainable AI (XAI)-based diagnostic model, standard classification performance metrics were computed. These metrics assess how well the CNN classifier distinguishes between normal and diseased cases across various medical image datasets such as pneumonia, tuberculosis, and skin cancer.

**Accuracy:**

Given TP, TN, FP, FN:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy measures the overall correctness of the model's predictions.
It represents the ratio of correctly classified images (both positive and negative) to the total number of test samples. In medical diagnosis, a higher accuracy indicates that the model is performing well in identifying both diseased and normal cases.

**Precision:**

Precision indicates the model's ability to correctly identify only the relevant (positive) cases among all samples predicted as positive.In the context of disease detection, precision shows how many of the X-rays predicted as diseased truly represent patients with the condition. High precision minimizes false alarms, reducing unnecessary clinical follow-ups.

$$\text{Precision} = \frac{TP}{TP + FP}$$

**Recall:**

Recall, also known as sensitivity, quantifies the model's capability to detect actual positive cases.In healthcare applications, a high recall is crucial since missing a positive (diseased) case could lead to severe consequences for the patient.This metric is prioritized when false negatives must be minimized.

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN}$$

**F1 Score:**

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score is the harmonic mean of precision and recall, balancing both metrics to provide a single performance indicator.
It is especially useful when the dataset is imbalanced—such as when healthy images far outnumber diseased samples—ensuring both sensitivity and specificity are equally valued.

**ROC- AUC :**

ROC–AUC evaluates the classifier's performance across different threshold settings by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR).

$$\mathbf{TPR} = \frac{TP}{TP+FN}$$

$$\mathbf{FPR} = \frac{FP}{FP+TN}$$

A higher AUC indicates better separability between diseased and normal cases.This means the model consistently assigns higher scores to positive samples than to negative ones, reflecting its diagnostic reliability.

These metrics are automatically computed after training and testing the CNN models (ResNet18, MobileNetV2, EfficientNetB0) across different datasets. The results are displayed in the web interface and visualized using confusion matrices and ROC curves. The F1-score and AUC values provide robust measures of model consistency and interpretability when integrated with Grad-CAM explanation.

### 3.3 Advantages and Limitations of the System

**Advantages**

1. **Explainability and Transparency:**Grad-CAM visualizations provide clear heatmaps that show *why* the model made a specific decision, increasing trust among medical professionals.
2. **Improved Clinical Interpretability:**Doctors can validate AI predictions by observing highlighted regions in X-rays or scans, making it easier to integrate the system into clinical workflows.
3. **High Diagnostic Accuracy:**The CNN model achieves strong performance across diseases such as Pneumonia, Tuberculosis, and Skin Cancer due to efficient feature extraction and robust training.
4. **Multi-Disease Capability:**The architecture is flexible and can be adapted to different medical datasets with minimal retraining effort.
5. **Automation of Image Analysis:**Reduces manual diagnostic workload and speeds up the detection process, which is crucial in time-sensitive healthcare situations.
6. **Scalability and Reusability:**The modular design allows easy addition of new models or datasets, enabling expansion into other disease categories.
7. **Ethical and Trustworthy AI:**Explainability ensures that the system follows medical accountability standards and aligns with ethical AI practices.

**Disadvantages / Limitations**

1. **Dependence on Dataset Quantity:**The system's performance is highly dependent on the availability of clean, labeled medical datasets. Biased or noisy data can lead to unreliable predictions.
2. **Limited Clinical Validation:**Although the model performs well on public datasets, real-world hospital data may yield varying results due to differences in imaging devices or patient demographics.
3. **Computational Requirements:**Deep learning training and Grad-CAM visualization require high-performance GPUs, which may limit deployment in low-resource environments.
4. **Interpretation Complexity:**While Grad-CAM provides visual explanations, the resulting heatmaps may still require expert interpretation—non-specialists could misread them.
5. **Black-Box Elements Remain:**Even with XAI integration, the internal representation of CNN layers is not fully interpretable, and some level of opacity remains.
6. **Risk of Overfitting:**Small or unbalanced datasets can lead the CNN to overfit, reducing its generalization capability on unseen data.
7. **Regulatory and Ethical Barriers:**Deployment in clinical settings requires medical certification and compliance with regulations, which can be time-consuming and complex.

## IV. ANALYSIS ON RESULTS

The results of the proposed diagnostic framework based on XAI have been explored using various medical image datasets related to chest X-rays, tuberculosis radiographs, and dermoscopic images. Using a CNN architecture like ResNet-18, deep learning representation learns to classify pathological images against normal images. In most cases, the model provides high accuracy and strong generalization capability across multiple datasets, thus showing its robustness in recognizing diverse disease patterns.

The quantitative results using the confusion matrix, precision, recall, and F1-score illustrate that the CNN trained in this work efficiently discriminates the cases into infected and non-infected ones, since most of the trials yielded a value greater than 90% for both precision and recall. The analysis of the ROC–AUC curve confirms the discriminative power of the model; it guarantees the reliability of the model's performance under variable conditions of image acquisition.

Qualitative analysis, using Grad-CAM visualizations, provides a better understanding of the model's decision process beyond raw accuracy. The heatmaps clearly bring forth clinically relevant regions within the input images,

such as lung opacities or lesion areas, corresponding to the features that influenced the model's prediction. It is interpretability that will help medical practitioners bridge the gap between automated diagnosis and human understanding. By showing which regions contribute most to each classification, Grad-CAM allows the validation of AI-generated decisions and helps prevent over-reliance on black-box models.

Generally, the analysis testifies that integration of CNN with XAI techniques not only provides high classification accuracy but also transparent and trustworthy output for clinical decision support. The demonstration of quantitative metrics along with model explainability is a justification that the proposed model is both technologically efficient and medically interpretable; thus, it serves the emerging need for responsible and accountable AI in health care.

## V.CONCLUSION

A fusion of XAI with CNN offers a very powerful and transparent approach to medical image analysis. With deep learning and interpretability techniques like Grad-CAM, the proposed system provides high diagnostic accuracy and gives visual explanations that enhance clinical trust in decision-making.

It highlights critical regions in X-rays and other medical scans that allow practitioners to understand and validate the AI predictions to bridge the gap between automated diagnosis and medical reasoning. This whole framework is explainable, hence enabling AI to act as an assistive tool but not a black-box system in order to generate greater confidence in real-world applications in healthcare.

In general, the results of this study ensure that explainable deep learning models are reliable, interpretable, and efficient diagnostic aids for safety and ethics in the adoption of AI in clinical environments.

## REFERENCES


[1] P. Purwonno, A. N. E. Wulandsari, and K. Nisa, "Explainable AI in Medical Imaging: Techniques, Applications, Challenges and Future Directions," *Procedia Computer Science*, 2024.

[2] K. Borys, Y. A. Schmitt, M. Nauta, C. Seifert, N. Kramer, C. M. Fedrich, and F. Nensa, "Explainable AI in Medical Imaging: An Overview for Clinical Practitioners—A Saliency-Based XAI Approach," *Frontiers in Artificial Intelligence*, vol. 7, pp. 112–124, 2024.

[3] R. Thakur, "Explainable AI: Developing Interpretable Deep Learning Models for Medical Diagnosis," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 2, 2024.

[4] A. Singh, S. Sengupta, and V. Lakshmi Narayanan, "Explainable Deep Learning Models in Medical Image Analysis," *Biomedical Signal Processing and Control*, vol. 93, 2025.

[5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.

[6] G. Litjens et al., "A Survey on Deep Learning in Medical Image Analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.

[7] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Springer, 2015.

[8] L. Montavon, W. Samek, and K. R. Müller, "Methods for Interpreting and Understanding Deep Neural Networks," *Digital Signal Processing*, vol. 73, pp. 1–15, 2018.

[9] A. Holzinger, G. Langs, H. Denk, K. Zatloukal, and H. Müller, "Causability and Explainability of Artificial Intelligence in Medicine," *WIREs Data Mining and Knowledge Discovery*, vol. 9, no. 4, 2019.

[10] M. Tjoa and C. Guan, "A Survey on Explainable Artificial Intelligence (XAI): Toward Medical XAI," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 11, pp. 4793–4813, 2021.

[11] D. Arrieta et al., "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges Toward Responsible AI," *Information Fusion*, vol. 58, pp. 82–115, 2020.

[12] H. Gupta, P. Sharma, and S. Jindal, "Explainable Convolutional Neural Networks for COVID-19 Diagnosis from Chest X-Rays," *Scientific Reports*, vol. 13, no. 11122, 2023.

[13] C. Lundberg and S. Lee, "A Unified Approach to Interpreting Model Predictions," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[14] N. Z. Mehmood and S. Qureshi, "AI-Enabled Diagnosis Systems Using Deep CNNs for Skin Lesion Classification," *IEEE Access*, vol. 11, pp. 31224–31237, 2023.

[15] F. Chollet, "Xception: Deep Learning with Depthwise Separable Convolutions," *Proceedings of the IEEE*

*Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1251–1258, 2017.